

Chapter 77

Managing Color for Visual Effects

This chapter discusses LUTs, color space conversions, and the value of compositing with linear gamma while previewing the image in the viewer using the gamma of your choice.

Contents

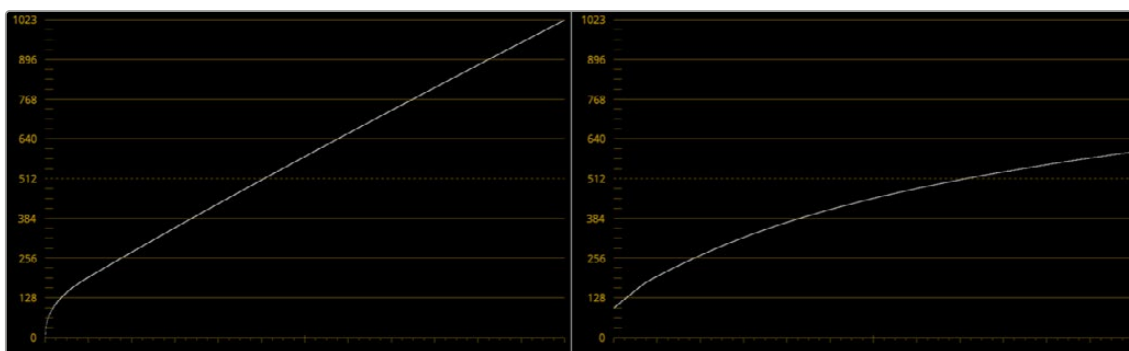
Color Management	1599
All Compositing Is Math.....	1600
Introducing Color Management in Fusion	1600
Converting to Linear Gamma.....	1601
Applying LUTs to a Viewer.....	1604
Using Resolve Color Management	1606
Using ACES Color Management in Resolve	1607
Using OCIO for ACES Color Management in Fusion	1608
Applying OCIO LUTs in the Viewer.....	1609

Color Management

The simplified goal of color management is to make sure that the image you see on your computer screen is what your audience sees when they view it on a television, cinema screen, or mobile device. For such a simple goal, problems arise that you, as the compositor, must manage. These problems begin with the fact that our eyes see luminance one way, and a computer display represents luminance differently.

Each capture device records images using a nonlinear tonal curve or gamma curve to compensate for this difference. Specifically, Rec. 709 HD gamma curves are designed so that when shown on HD displays, the images have built-in compensation for the display. The result is that HD images on HD displays appear normal to us.

Digital cinema cameras have taken the concept of gamma curves further. They use gamma curves as a way to maximize the bit depth of an image and store a wider dynamic range. Digital cinema cameras' gamma curve (often collectively referred to as *log gamma*), give more attention to the darker mid-tones where the human eye is most sensitive. This allows them to save images with brighter highlights and more detail in shadows.



A Rec. 709 HD gamma curve (left) and a nonlinear, or log gamma, curve (right)

The problem is that these images do not look normal on any monitor. Clips recorded with a log gamma curve typically have a low contrast, low saturated appearance when viewed on an sRGB computer display or Rec. 709 HD video monitor. This problem is easy to fix using a LookUp Table, or LUT. A LUT is a form of gamma and color correction applied to the viewer to normalize how the image is displayed on your screen.



A clip displayed with a nonlinear, log gamma curve (left) and corrected in the viewer using a LUT (right)

All Compositing Is Math

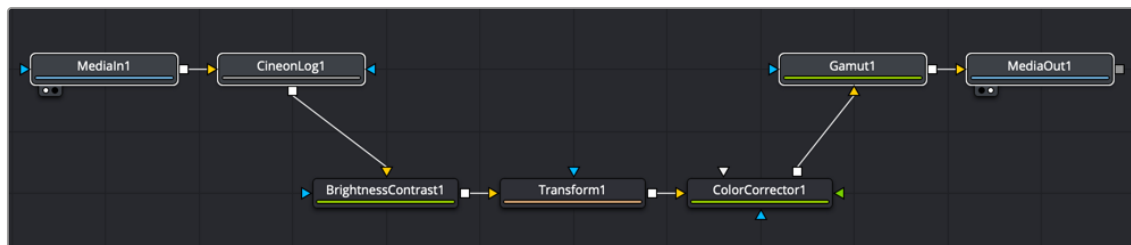
The LUT applied to the viewer only solves the problem in the viewer. Now we come to the larger problem. The image data is still using a log gamma curve. Fusion, and every other image-processing application, operates with the assumption that the image data has linear gamma. The image-processing filters you apply to images use standard math functions, like $1 + 1 = 2$. Common operations such as those that add pixels like Brightness, or divide pixels (a.k.a. “unpremultiply”), or composite modes that include multiplication such as “screen,” and many other compositing tasks assume that $1 + 1$ always equals 2. In other words, if you perform an operation that doubles the amount of brightness, then every pixel should be twice as bright. However, if you are starting with a nonlinear gamma curve, pixels are not being adjusted linearly, so some pixels might end up 1.2 x as bright, 1.7 x as bright, or 2.4 x as bright. Now the math is $1 + 1 = 3$. The further your images are from linear gamma, the more pronounced the math error. A Rec. 709 HD clip shows less error than a log gamma clip from a digital cinema camera. However, an error is still an error, and the more compositing operations you perform on the image, the more the error is compounded.

You can see a more practical example when you apply filtering effects, such as a blur, to an image with any gamma setting. The image probably looks fine. However, if you convert the image to a linear gamma first and then apply the blur, the images (especially those with extremely bright areas) are processed with greater accuracy, and you should notice a different and superior result.

The answer to these problems is to manage your color before compositing.

Introducing Color Management in Fusion

Images loaded into Fusion by default are not color managed. The image is displayed directly from the file to the viewer without any interpretation or conversion. However, Fusion includes nodes that convert the output of each image to linear gamma at the beginning of your composite. The same nodes can convert from linear back to your desired output gamma at the end of your composite, just prior to the Saver or MediaOut node.



A log clip converted to linear and then converted for output

To manually set up a linear gamma workflow in Fusion:

- 1 Use a Gamut or CineonLog node after all MediaIn or Loader nodes to convert them to linear.
- 2 Apply a GAMUT View LUT to the viewers to correct the display of a linear image to sRGB or Rec. 709.
- 3 Before a Saver or MediaOut node, insert a Gamut or CineonLog node to convert from linear to your target output format.

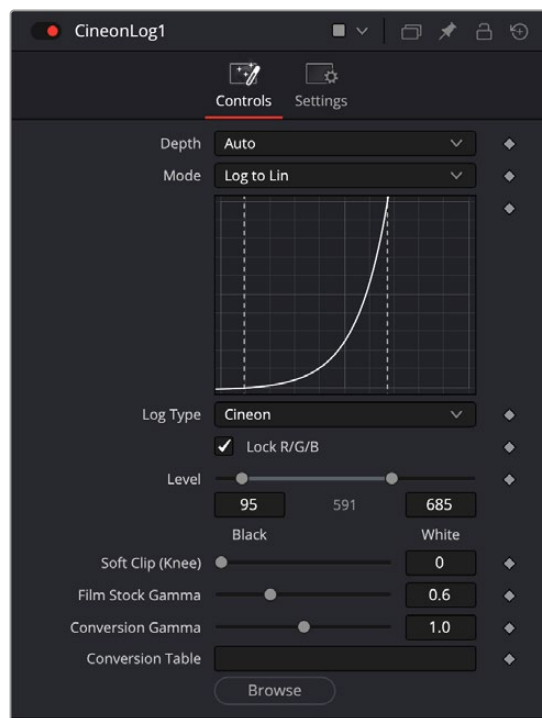
Converting to Linear Gamma

Whether an image comes from the Edit page in DaVinci Resolve, or from a Loader in Fusion Studio, the color and gamma are read directly into Fusion, with no modification. For some simple operations on sRGB or Rec 709 clips, this may be fine, but it's not always the ideal way to work, especially for log-encoded media. The ideal way to work with log-encoded media is to convert images to linear gamma, since the majority of image-processing operations in Fusion expect gamma to be linear and will produce superior results.

TIP: 3D rendered CGI images are often generated as EXR files with linear gamma, and converting them is not necessary. However, you should check your specific files to make sure they are using linear gamma.

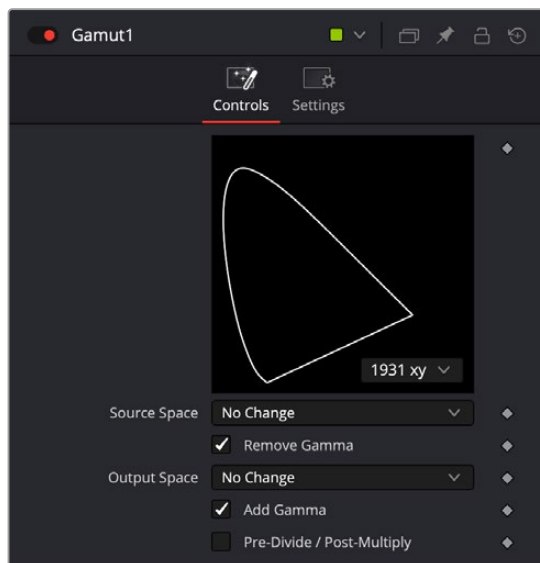
Fusion includes several kinds of nodes you can use to convert the image out of each MediaIn or Loader node to linear gamma at the beginning of your composite, and then convert from linear back to your desired output gamma at the end of your composite. These include:

- **CineonLog node:** The CineonLog node, found in the Film category of the Effects Library, performs a conversion from any of the formats in the Log Type menu to linear, and also reverses the process, adding log gamma back to a clip. This is most often used for images coming from common digital cinema cameras like BlackMagic Design, Arri, or Red. The CineonLog node is added directly after a MediaIn or Loader node. The Mode menu chooses the direction of the conversion to or from linear.



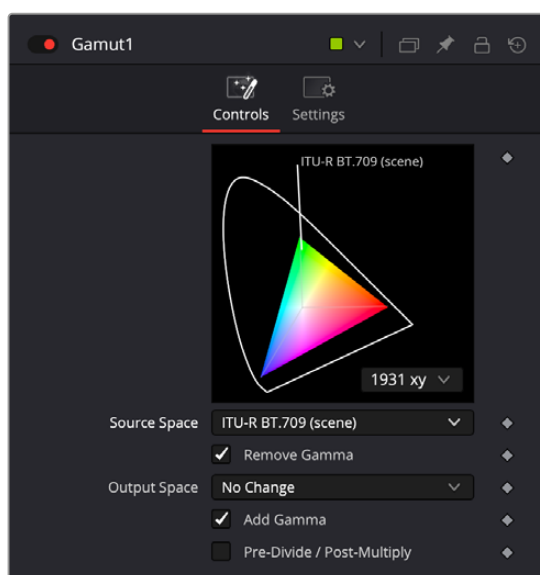
Add a CineonLog tool to convert log gamma curves to linear and vice versa.

- **Gamut node:** The Gamut node, found in the Color category of the Effects Library, lets you perform linear conversions based on color space. This node converts to linear or from linear and is often inserted after a MediaIn or Loader node or just before a MediaOut or Saver node. Depending on where you insert the node, you either choose from the Source Space controls or the Output Space controls.



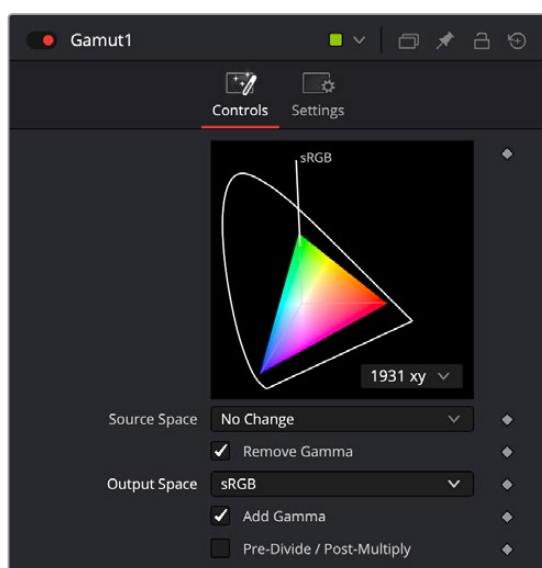
Add a Gamut tool to convert gamma curves to linear based on color space.

When converting media to linear gamma, set the Source Space menu to the color space of your source material. For instance, if your media is full 1080 HD ProRes, then choose ITU-R BT.709 (scene) for gamma of 2.4. Then, enable the Remove Gamma checkbox if it isn't already enabled, to use linear gamma.



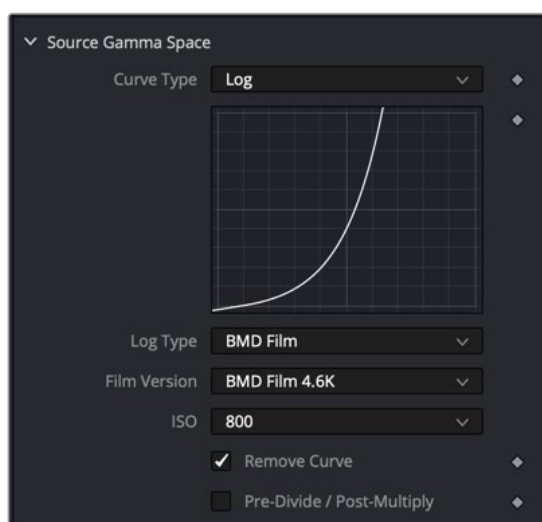
Source Space is used to convert to linear gamma.

When converting from linear gamma for output, you insert the Gamut node before your output node, which is a Saver in Fusion Studio or a MediaOut node in DaVinci Resolve's Fusion page. Make sure the Source Space menu is set to No Change, and set the Output Space to your output color space. For instance, if your desired output is full 1080 HD, then choose either sRGB or ITU-R BT.709 (scene) for gamma of 2.4. Then, enable the Add Gamma checkbox if it isn't already enabled, to format the output of the Gamut node for your final output.



Output Space is used to convert from linear gamma.

- MediaIn and Loader nodes:** MediaIn and Loader nodes have Source Gamma Space controls in the Inspector that let you identify and remove the gamma curve without the need to add another node. If your files include gamma curve metadata like RAW files, the Auto setting for the Curve Type drop-down menu reads the metadata and uses it when removing the gamma curve. When using intermediate files or files that do not include gamma curve metadata, you can choose either a log gamma curve by choosing Log from the Curve Type menu or a specific color space using the Space option from the menu. Clicking the Remove Curve checkbox then removes the gamma curve, converting the image to linear gamma.



MediaIn and Loader nodes include a Remove Curve checkbox in the Inspector.

- FileLUT node:** The FileLUT node, found in the LUT category of the Effects Library, lets you do a conversion using any LUT you want, giving you the option to manually load LUTs in the ALUT3, ITX, 3DL, or CUBE format to perform a gamma and gamut conversion. Although LUTs are very commonly placed at the end of a node tree for final rendering, you'll get more accurate gamma and color space conversions using the Gamut and CineonLog nodes to transform your MediaIn and Loader nodes into linear.

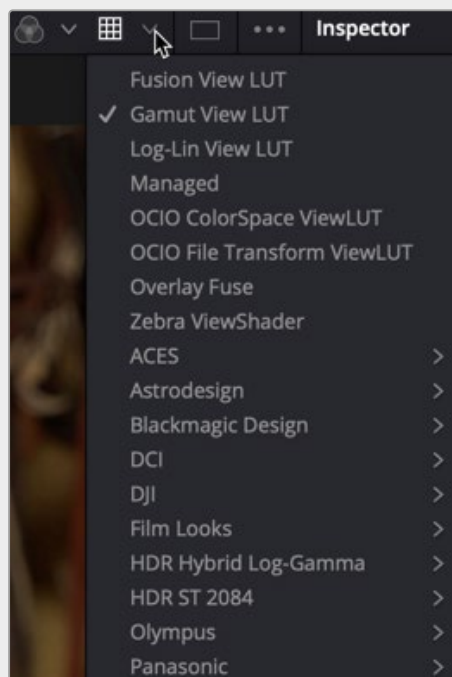
Applying LUTs to a Viewer

Images converted to a linear gamma don't look correct. They usually look very dark, with extremely bright highlights and oversaturated colors. Happily, even though the image may appear to be incorrect, the fact that Fusion can work entirely with floating-point color data means that you're not actually clipping or losing any image data. It just looks completely wrong when viewing the linear state of your image data directly.



A clip displayed with a nonlinear, log gamma curve (left) and the clip transformed to linear gamma (right)

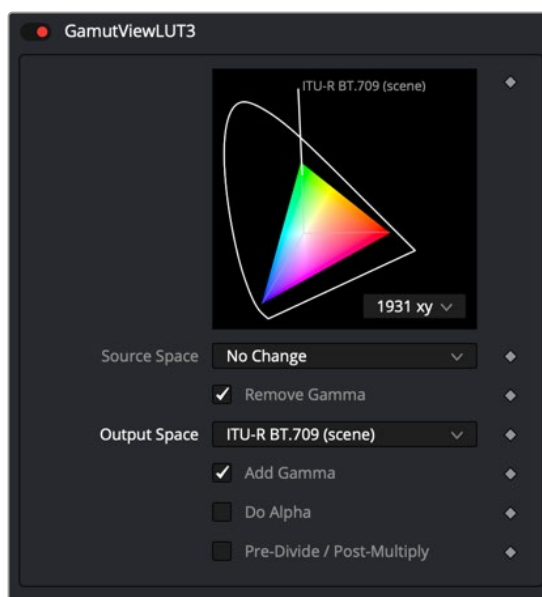
It would be impossible to work if you couldn't view the image as it's supposed to appear within the final gamut and gamma you'll be outputting. For this reason, each viewer has a LUT menu that lets you enable a "preview" color space and/or gamma conversion, while the node tree is processing correctly in linear gamma.



Apply a Gamut View LUT to preview the image in your intended output color space.

To preview the images in the viewer using sRGB or Rec. 709 color space:

- 1 Enable the LUT button above the viewer.
- 2 From the Viewer LUT drop-down menu, choose either a Gamut View LUT, or a LUT from the VFX IO category that transforms linear to Rec. 709 or sRGB.
- 3 If you choose the Gamut View LUT, then choose Edit from the bottom of the LUT menu to configure the LUT.
- 4 In the LUT Editor, set the Output Space to the target color space you want.
- 5 Enable the Add Gamma checkbox to apply the gamma curve based on the selected color space.



Gamut View LUT Output Space set to Rec. 709 for HD with the Add Gamma checkbox enabled

TIP: If your monitor is calibrated differently, you need to select a LUT that matches your calibration.

Whether you use the Gamut View LUT or a LUT for your specific monitor calibration, you can save the viewer setup as the default.

To Save the Gamut LUT setup as the default viewer setup:

- Right-click in the viewer, and then choose Settings > Save Defaults.

For every comp, the viewer will now be preconfigured based on the saved defaults.

For more information on Viewer LUTs, see *Chapter 69, "Using Viewers."* in the DaVinci Resolve Reference Manual or *Chapter 7* in the Fusion Reference Manual.

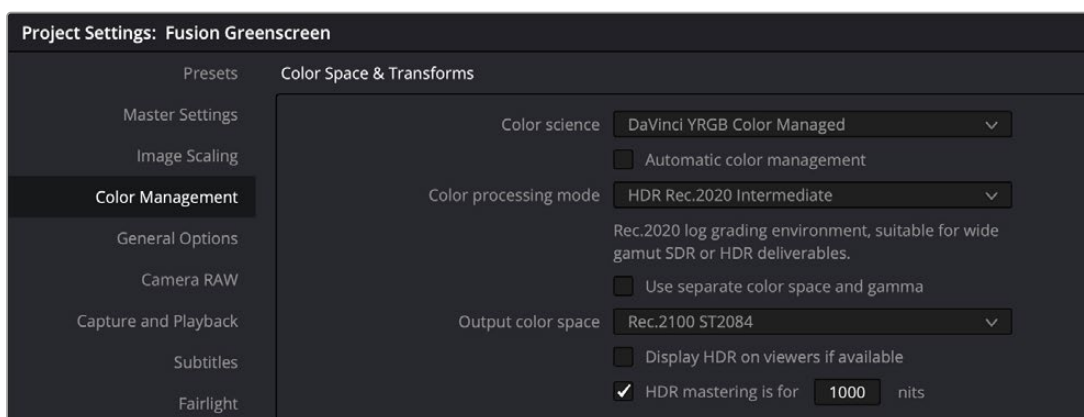
Using Resolve Color Management

If you're using the Fusion page within Davinci Resolve, you have the option of enabling DaVinci Resolve's scene-referred color management, instead of inserting Gamut and CineonLog nodes. When DaVinci YRGB Color Managed (RCM) is enabled, the color of MediaIn nodes in the Fusion page is handled differently. The RCM automatically determines the input color spaces of all files used for all MediaIn nodes in the Fusion page, which then automatically get converted to linear gamma. The MediaOut node then gets converted back into the Color Processing mode to get graded in the Color page or further edited in the Edit page Timeline.

Despite the seeming complexity of color management, using RCM is actually simple. In essence, all you have to do is (A) turn on RCM, and (B) choose the Color Processing mode and Output Color Space combination you want to use.

To enable Resolve Color Management:

- 1 Open the Color Management panel of the Project Settings.
- 2 Choose DaVinci YRGB Color Managed from the Color Science drop-down menu.
- 3 Check the Automatic color management box for a simplified selection, or un-check the box to adjust the Color Processing mode and Output color spaces manually.
- 4 Set the desired options for the Color Processing mode and Output Color Space RCM settings.



When DaVinci YRGB Color Managed is enabled, Timeline color space is used for all MediaIn nodes in the Fusion page.

To override the input color space for differently recorded clips in the Media Pool:

- 1 Enable DaVinci YRGB Color Management as explained above.
- 2 Save and close the Settings dialog.
- 3 In the Media Pool, select the clip or clips you want to assign a new Input Color space.
- 4 Right-click one of the selected clips.
- 5 Choose the Input Color Space that corresponds to those clips from the contextual menu.

Using RCM eliminates a few steps, since the input color space math used to transform the source preserves all wide-latitude image data, making highlights easily retrievable without any extra steps. With RCM enabled, there is no need to insert CineonLog or Gamut nodes while in the Fusion page. The transforms from and to linear are done automatically based on the RCM settings. Switching to the Fusion page converts the images to linear and enables the LUT button in the viewers with the Managed LUT selected. The Managed LUT uses the RCM settings to take a linear image and display it based on the RCM output color space.

For more information on Resolve Color Management, see *Chapter 9, “Data Levels, Color Management, and ACES.”*

Using ACES Color Management in Resolve

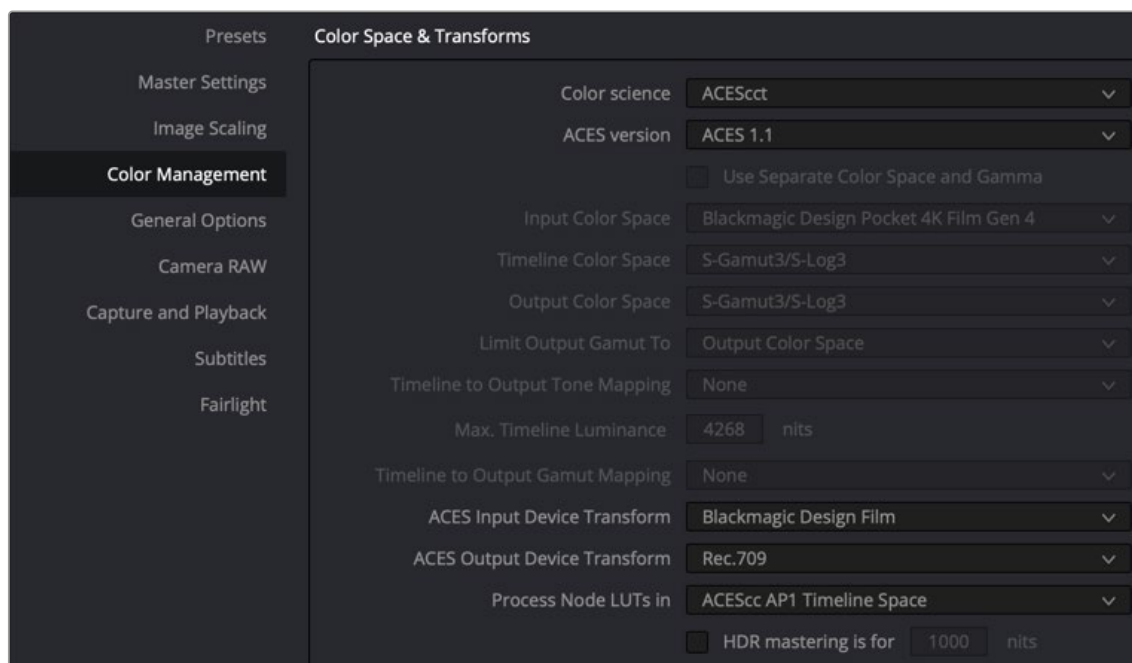
The ACES (Academy Color Encoding Specification) color space is another standard for managing color throughout an entire production. It’s designed to make start-to-finish, scene-referred color management a reality for digital cinema workflows. Just like DaVinci’s RCM, ACES makes it easier to extract high-precision, wide-latitude image data from raw camera formats, to preserve high-quality image data from acquisition through the color grading process and to output high-quality data for broadcast viewing, film printing, or digital cinema encoding.

ACES works by assigning an IDT (Input Device Transform) to every camera and acquisition device. The IDT specifies how media from that device is converted into the ACES color space. At the end of the pipeline, an ODT (Output Device Transform) is applied to convert the image data from ACES color space into the gamut of your final output.

Similar to setting up RCM, DaVinci Resolve’s color management project settings can be configured for ACES, which carries through the Edit, Fusion, and Color pages.

NOTE: When using Fusion Studio, the OpenColorIO (OCIO) framework is used for ACES color management.

The Color Science drop-down menu in the Color Management panel of the Project Settings is used to set up the ACES color management in DaVinci Resolve.



When ACES is enabled, IDT and ODT are used to identify input and output devices.

- **Color Science:** Using this drop-down menu, you can choose either ACEScct or ACEScc color science. This is primarily a personal preference since they are mostly identical, but the shadows respond differently to grading operations. In the Fusion page, images are automatically converted to linear, so whoever does the grading has more of a reason to choose one or the other.

ACEScc: Choose ACEScc color science to apply a standard Cineon style log encoding to the ACES data before it is processed by DaVinci Resolve.

ACEScct: This variation of ACEScc adds a roll-off at the toe of the image to make color correction lift operations “feel” more like they do with film scans and LogC encoded images.

- **ACES Version:** When you’ve chosen one of the ACES color science options, this menu becomes available to let you choose which version of ACES you want to use.
- **ACES Input Device Transform:** This menu lets you choose which IDT (Input Device Transform) to use for the dominant media format in use.
- **ACES Output Device Transform:** This menu lets you choose an ODT (Output Device Transform) with which to transform the image data to match your required deliverable.
- **Process Node LUTs In:** This menu lets you choose how you want to process LUTs in the Color page and does not affect the Fusion page.
- **Disable tone mapping for Fusion conversion:** Checking this box will remove any tone mapping from the ACES color management.

For more information on ACES within DaVinci Resolve, see *Chapter 9, “Data Levels, Color Management, and ACES.”*

Using OCIO for ACES Color Management in Fusion

When using Fusion Studio or not using color management in DaVinci Resolve, you have the option to use OpenColorIO nodes in Fusion to composite in an ACES color space.

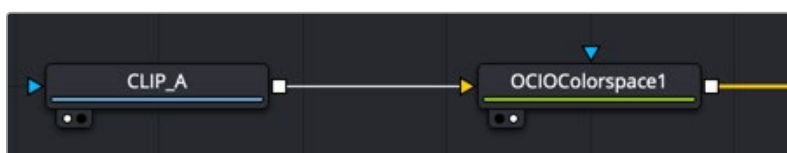
OpenColorIO (OCIO) is an open-source color management framework for visual effects and computer animation. OCIO is compatible with the Academy Color Encoding Specification (ACES). Three OCIO nodes located in the Color category of the Effects Library allow you to use OCIO color space transforms in Fusion.

OCIO CDL Transform node allows you to create, save, load, and apply a Color Decision List (CDL) grade.

OCIO Color Space allows sophisticated color space conversions based on an OCIO config file.

OCIO File Transform allows you to load and apply a variety of LookUp Tables (LUTs).

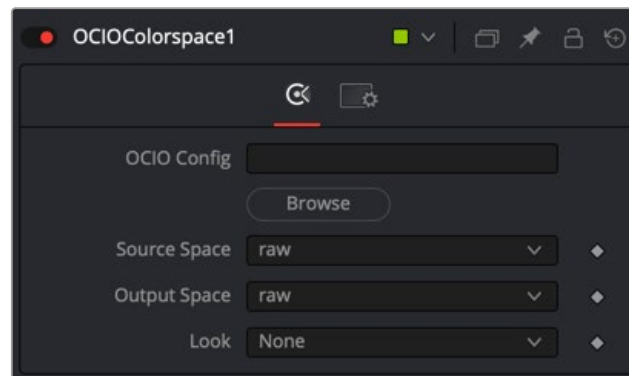
Using OCIO for converting MediaIn or Loader nodes to linear gamma is based on the OCIO Color Space node. Placing the OCIO Color Space node directly after a Loader (or MediaIn in DaVinci Resolve) displays the OCIO Source and Output controls in the Inspector.



OCIO Color Space nodes can be used to work in an ACES color managed environment.

Within the Inspector for OCIO Color Space node, Fusion includes default Source and Output transforms for standard color spaces. However, to use the full OCIO standard, you'll need to download and install the OCIO config file. You can download the config file from the OCIO website. <https://opencolorio.org>

Clicking the Browse button in the Inspector will allow you to navigate to the downloaded config file. From the download, locate the ACES 1.0.3 or later folder and select the file config.ocio.



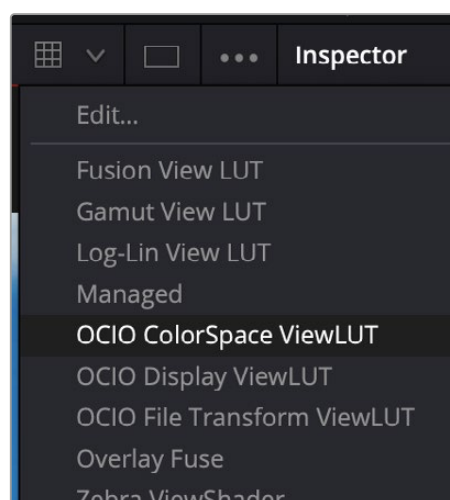
Source and Output menus are populated based on the config.ocio file that you download.

The Source menu is used to choose the color profile for your Loader or MediaIn node. The default raw setting shows an unaltered image, essentially applying no color management to the clip. The selection you make from the menu is based on the recording profile of your media.

The Output menu is set based on your deliverables. When working in Fusion Studio, typically the Output selected is ACEScg, to work in a scene linear space.

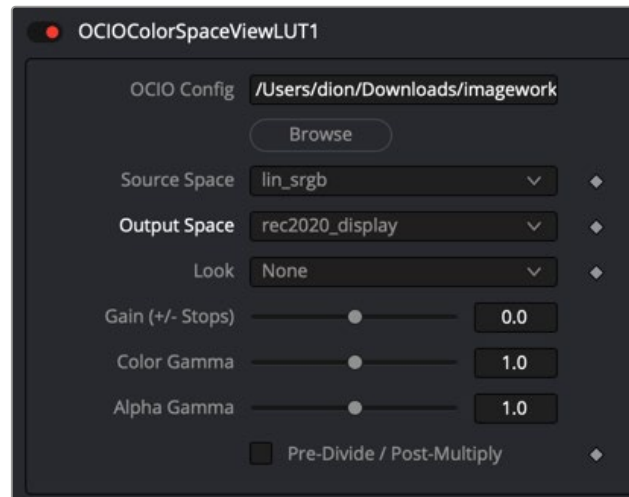
Applying OCIO LUTs in the Viewer

The viewer also includes OCIO View LUTs to calibrate the viewers. Once the OCIO Color Space View LUT is selected from the LUT menu above the viewer, choosing Edit for the same menu opens a dialog where you can load the OCIO config file.



The OCIO Color Space View LUT is located in the Viewer LUT menu in Fusion Studio.

By default, the same standard options are available in the View LUT. However, clicking the Browse button allows you to load the same config file you loaded into the OCIO Color Space node. Once loaded, all the expanded OCIO options are available. If you selected the OCIO Color Space node to output ACEScg, you use the OCIO View LUT to go from a source setting of linear sRGB to an output setting of sRGB or Rec. 709 in most cases.



The OCIO Color Space View LUT dialog is used to configure the viewer when using the OCIO Color Space node in the Node Editor.

TIP: If your monitor is calibrated differently, you will need to select a LUT that matches your calibration.

To set the LUT using OCIO:

- 1 Click the LUT menu and choose the OCIO Color Space View LUT.
- 2 From the same menu, select Edit.
- 3 In the View LUT editor that opens, set the source's color space to lin sRGB
- 4 Set the output space to sRGB or REC 709, assuming you are viewing on a standard computer monitor. You now see a normalized image in the viewer, but all color operations will be on linear images.

Whether you use the OCIO Color Space LUT or a LUT for your specific monitor calibration, you can save the viewer setup as the default.

To save the OCIO ColorSpace LUT setup as the default viewer setup:

- Right-click in the viewer, and then choose Settings > Save Defaults. Now, for every comp, the viewer is preconfigured based on the saved defaults.